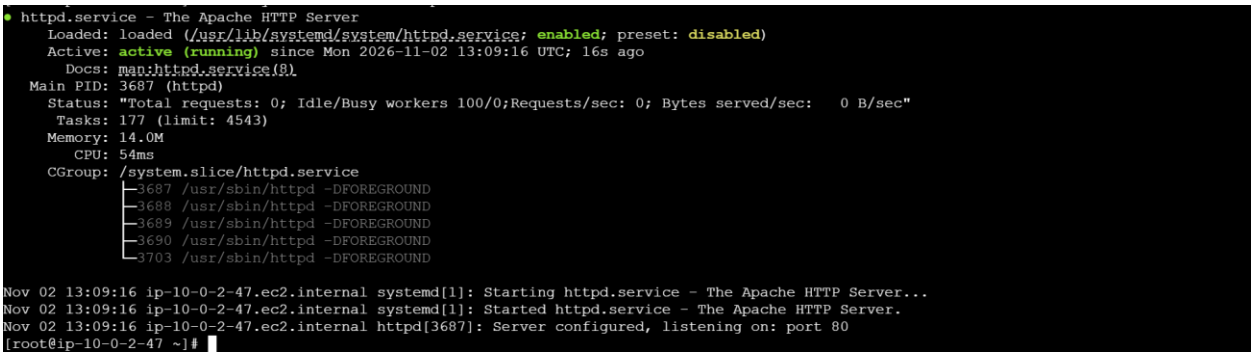
Terraform – 5

Create an EC2 instance with Apache (`httpd`)



terraform { required_providers { aws = { source = "hashicorp/aws" version = "~> 6.0" } } }

provider "aws" { region = "us-east-1" }

data "aws_vpc" "existing" { id = "vpc-019405835908e8978" }

data "aws_subnets" "public" { filter { name = "vpc-id" values = [data.aws_vpc.existing.id] }

filter { name = "tag:Name" values = ["*public*"] } }

resource "aws_security_group" "web_sg" { name = "terraform-05-web-sg" vpc_id = data.aws_vpc.existing.id

ingress { from_port = 80 to_port = 80 protocol = "tcp" cidr_blocks = ["0.0.0.0/0"] }

egress { from_port = 0 to_port = 0 protocol = "-1" cidr_blocks = ["0.0.0.0/0"] } }

resource "aws_instance" "web_server" { ami = "ami-0c230065b91270bc6" instance_type = "t3.micro" subnet_id = data.aws_subnets.public.ids[0] vpc_security_group_ids = [aws_security_group.web_sg.id]

user_data = <<-EOF #!/bin/bash dnf install -y httpd systemctl enable --now httpd echo "Hello! Apache installed using Terraform." > /var/www/html/index.html EOF

tags = { Name = "Terraform-05-WebServer" } }

Enter this in the powershell

```
PS C:\Users\india\OneDrive\Desktop\terraform-05-ec2> terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.71.0...
- Installed hashicorp/aws v6.71.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

Terrform will create this;

Resource	Details
Ec2 instance	t3.micro
AMI	Amazon Linux 2023
VPC	Existing VPC
Subnet	subnet-04a0839a233b0012c
Security group	terraform-05-web-sg
Inbound rule	HTTP port 80

```
  Enter a value: yes

aws_security_group.web_sg: Creating...
aws_security_group.web_sg: Creation complete after 6s [id=sg-06b9604c21603db06]
aws_instance.web_server: Creating...
aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Creation complete after 16s [id=i-06ddc8b0944a9d9c7]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\india\OneDrive\Desktop\terraform-05-ec2>
```

```
-instance-ids i-06ddc8b0944a9d9c7 --query "Reservations[0].Instances[0].{State:St
ate.Name,Subnet:SubnetId,PublicIP:PublicIpAddress,PrivateIP:PrivateIpAddress,Publ
icIPAssociation:NetworkInterfaces[0].Association.PublicIp}" --output table
-----------------------------------------------------
|                 DescribeInstances                 |
+----------------------+----------------------------+
|  PrivateIP           |  10.0.2.98                 |
|  PublicIP            |  None                      |
|  PublicIPAssociation |  None                      |
|  State               |  running                   |
|  Subnet              |  subnet-04a0839a233b0012c  |
+----------------------+----------------------------+
```

aws_instance.web_server: Destroying... [id=i-06ddc8b0944a9d9c7]

aws_instance.web_server: Still destroying... [id=i-06ddc8b0944a9d9c7, 00m10s elapsed]

aws_instance.web_server: Still destroying... [id=i-06ddc8b0944a9d9c7, 00m20s elapsed]

aws_instance.web_server: Still destroying... [id=i-06ddc8b0944a9d9c7, 00m30s elapsed]

aws_instance.web_server: Destruction complete after 32s

aws_security_group.web_sg: Modifying... [id=sg-06b9604c21603db06]

aws_security_group.web_sg: Modifications complete after 2s [id=sg-06b9604c21603db06]

aws_instance.web_server: Creating...

aws_instance.web_server: Still creating... [00m10s elapsed]

aws_instance.web_server: Creation complete after 16s [id=i-051ada5ae40d9aff3]

Apply complete! Resources: 1 added, 1 changed, 1 destroyed.

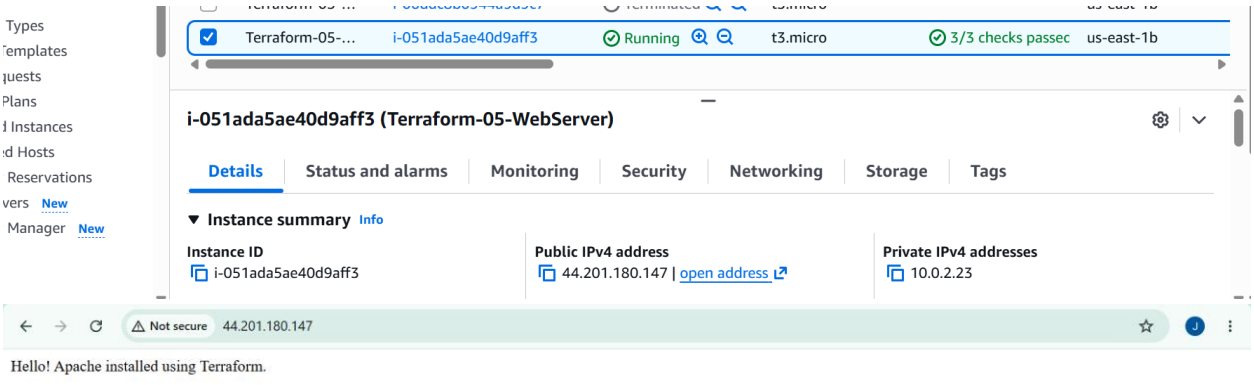
Outputs:

instance_id = "i-051ada5ae40d9aff3"

public_ip = "44.201.180.147"

EC2 successfully created!

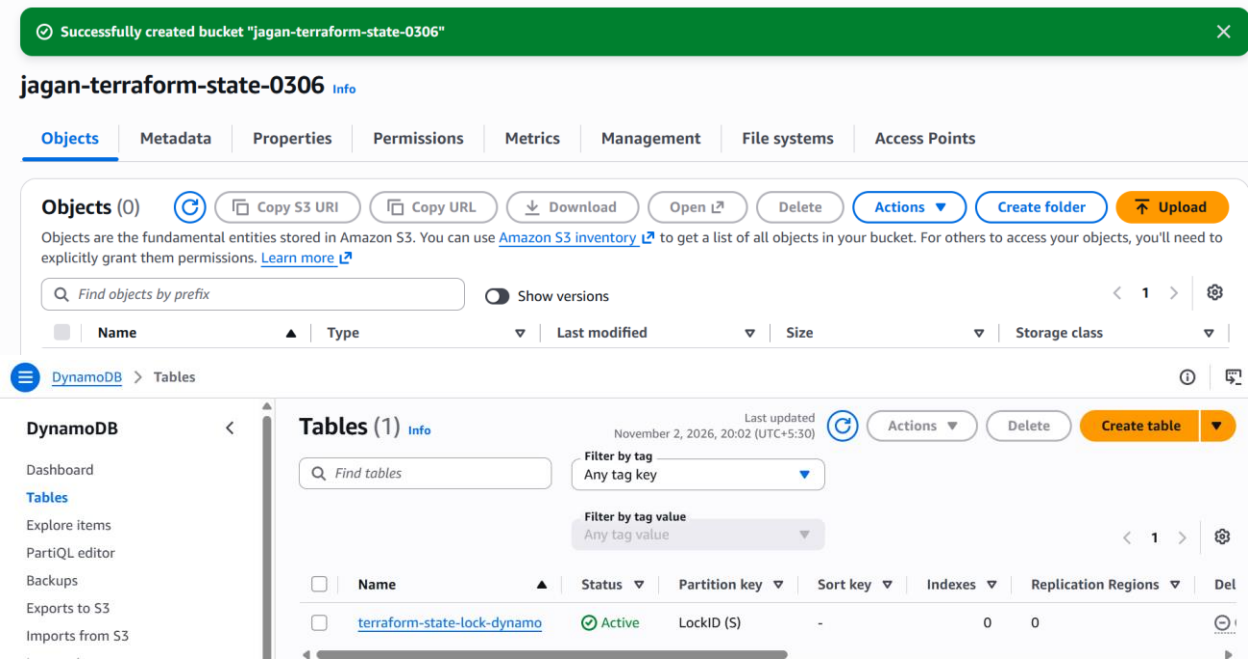
Excellent! Everything worked. Terraform replaced the old instance and created a new one with a public IP.



completed

- Create EC2 instance using Terraform.
- Install Apache (`httpd`) using Terraform `user_data`.
- Configure security group for HTTP port 80.
- Assign public IP.
- Verify Apache through browser.

Create an S3 bucket with the name `jagan-terraform-state-0306` in the US East (N. Virginia) – `us-east-1` region. Keep Object Ownership set to ACLs disabled, ensure Block all public access is enabled, and enable Bucket Versioning. These settings will help keep your Terraform state file private and maintain previous versions for recovery.



Configure S3 backend in Terraform

Open your `main.tf`.

At the very top, add this backend block inside the existing `terraform` block, before `required_providers`:

```
terraform {

 backend "s3" {

   bucket       = "jagan-terraform-state-0306"

   key          = "terraform-05/terraform.tfstate"

   region       = "us-east-1"

   dynamodb_table = "terraform-state-lock-dynamo"

   encrypt      = true
```

```
}

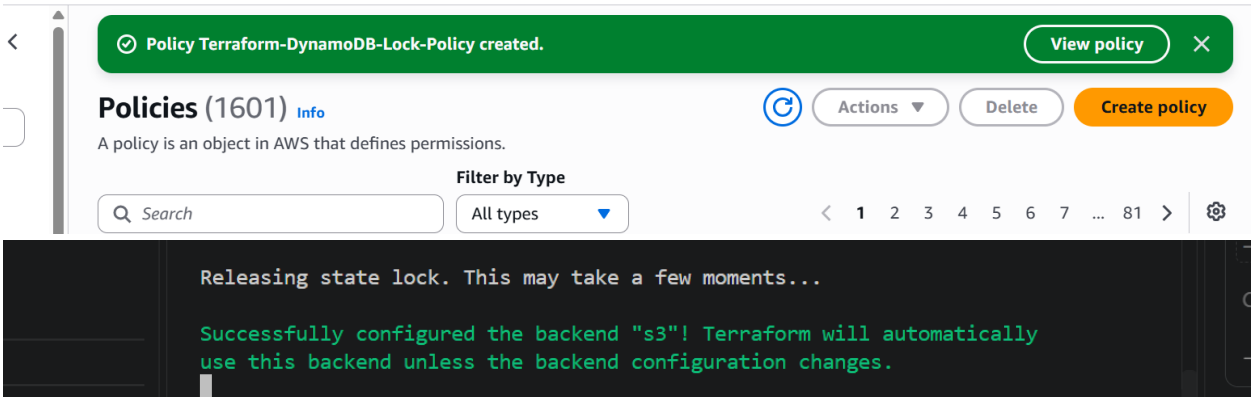

required_providers {
  aws = {
    source  = "hashicorp/aws"
    version = "~> 6.0"
  }
 }
}
```

```
public_ip = "44.201.180.147"
PS C:\Users\india\OneDrive\Desktop\terraform-05-ec2> terraform init -migrate-stat
e
```

Your S3 backend configuration is correct. The problem is an IAM permission issue.

Terraform is trying to access DynamoDB to acquire a state lock, but your IAM user `Iam-task-user` doesn't have the required permissions.





completed

- **Created EC2 instance using Terraform.**
- **Installed Apache using `user_data`.**
- **Created S3 bucket.**
- **Enabled S3 bucket versioning.**
- **Created DynamoDB table.**
- **Configured DynamoDB state locking.**
- **Migrated Terraform state to S3.**

Verify the remote state

Go to AWS Console → S3 → `jagan-terraform-state-0306`

